

FedPoisonTTP: A Threat Model and Poisoning Attack for Federated Test-Time Personalization

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Abstract

*Test-time personalization in federated learning enables models at clients to adjust online to local domain shifts, enhancing robustness and personalization in deployment. Yet, existing federated learning work largely overlooks the security risks that arise when local adaptation occurs at test time. Heterogeneous domain arrivals, diverse adaptation algorithms, and limited cross-client visibility create vulnerabilities where compromised participants can craft poisoned inputs and submit adversarial updates that undermine both global and per-client performance. To address this threat, we introduce **FedPoisonTTP**, a realistic grey-box attack framework that explores test-time data poisoning in the federated adaptation setting. FedPoisonTTP distills a surrogate model from adversarial queries, synthesizes in-distribution poisons using feature-consistency, and optimizes attack objectives to generate high-entropy or class-confident poisons that evade common adaptation filters. These poisons are injected during local adaptation and spread through collaborative updates, leading to broad degradation. Extensive experiments on corrupted vision benchmarks show that compromised participants can substantially diminish overall test-time performance.*

1. Introduction

Smartphones, browsers, and a growing array of internet-connected sensors and Web-of-Things (WoT) gadgets generate a large share of global network traffic and continuously produce diverse, user-generated content and telemetry. This stream of heterogeneous data is leveraged by machine-learning systems that power many web services, personalized recommendation [9, 39], social feeds [40], and content moderation [2, 23] among them. Because these models rely on information produced or held on users' devices, they create important privacy and data-control chal-

lenges for both users and service operators [1, 12, 42]. In this regard, Federated Learning (FL) has emerged as a practical paradigm for training machine learning models across decentralized clients while preserving data locality and user privacy [6, 21, 30]. Additionally, in many real-world deployments the data seen by each client is nonstationary: clients encounter domain shifts over time due to changing sensors, environments, user behaviors, or acquisition conditions. Hence, Test-Time Adaptation (TTA) is necessary to adapt the models in real time to the distribution shifts of the incoming data [32]. To maintain performance in the presence of such shifts in collaborative learning, recent work [24] has explored *federated test-time adaptation* (FTTA), a hybrid workflow in which a shared global model is periodically aggregated on the server while each client performs lightweight, unsupervised adaptation on its own incoming unlabeled stream at test time. By combining federation for shared representation learning with per-client test-time adaptation (TTA) routines (for example, batch-normalization (BN) adaptation, entropy minimization, or self-supervised consistency updates) [13, 19, 32], FTFA provides a practical path to personalized, robust inference without accessing or transmitting raw client data.

Despite these practical advantages, both federated learning and test-time adaptation are vulnerable to adversarial attacks [4, 14, 25, 34]. Federated systems are vulnerable to classical and model-poisoning attacks: compromised clients can send malicious updates to skew the global model, leak information, or induce targeted misclassification [3]. Many defense strategies, like robust aggregation, clipping, differential privacy, etc., reduce but do not fully eliminate these risks, and they often trade off utility [17, 22, 28]. Meanwhile, TTA algorithms rely on online unsupervised objectives such as entropy minimization or dynamic BN updates and therefore, can be manipulated by carefully crafted inputs that steer the adaptation process toward harmful local minima [10]. Prior work in the centralized setup has demonstrated that an adversary with strong access (white-box, or with knowledge of a held-out

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benign validation set) can craft per-example perturbations that derail TTA [8, 34]. However, these threat models do not capture a realistic federated deployment: an adversarial client in FL typically cannot observe other clients’ data or per-client updates, interacts with the server only via broadcast/aggregation, and participates intermittently with a limited budget for poisoning.

Bridging these two literatures, we study how a constrained adversary operating as one (or a few) malicious participants in an FTTA system can mount a highly effective and stealthy poisoning attack. The key challenge is *realism*: the attacker must succeed despite only seeing the global model broadcasts, lacking access to other clients’ local data or updates, and being limited in how many and which rounds they can influence. Drawing inspiration from recent single-node online poisoning approaches [25, 34], we develop a practical federated attack that rests on three core ideas. First, the adversary builds and continually refines a lightweight *surrogate aggregator* that predicts the post-aggregation model dynamics by combining history-based extrapolation of past global updates with posterior distillation on locally observed responses. Second, the adversary crafts *in-distribution* poisoned examples whose intermediate feature statistics are constrained to match those of a benign local pool via a feature-distribution regularizer; this encourages the harmful perturbations to transfer across heterogeneous client domains and to evade simple detection heuristics. Third, the attack optimizes TTA-aware objectives that explicitly target common adaptation rules deployed by clients (for example, shifting BN channel statistics to mislead BN-adaptive TTA, or shaping entropy/confidence in a class-balanced way to subvert entropy-minimizing updates). All components are designed to preserve stealth: per-example perturbation bounds, class balance, and upload-norm clipping are enforced so that the poisoned updates resemble ordinary client updates and evade naïve aggregation defenses.

This paper makes the following contributions:

- To the best of our knowledge, this is the first work to study adversarial (poisoning) attacks in federated test-time adaptation. We formalize the problem and introduce a realistic threat model where the adversary is a limited federated client that cannot access other clients’ data or local models and only observes global model broadcasts.
- We propose a practical attack framework that combines history-based surrogate aggregation, posterior distillation, and a feature-distribution regularizer to craft in-distribution poisoned examples that transfer across heterogeneous client domains. The attack is TTA-aware: it explicitly targets BN-based adaptation and entropy-driven updates that are commonly used in FTTA.
- Through a comprehensive empirical evaluation on federated domain-shift benchmarks and a variety of client

TTA methods, we demonstrate that our attack produces substantial and persistent degradation across many honest clients while remaining difficult to detect by basic defensive measures. We further analyze key design choices to understand which components most significantly contribute to the attack’s effectiveness.

2. Related Work

2.1. Test-time adaptation

Test-time adaptation (TTA) techniques adapt pretrained models at inference to mitigate distribution shift without access to source labels. Early practical methods showed that updating normalization statistics or simple self-supervised objectives at test time yields substantial improvements [16, 29]. *Tent* (entropy minimization with BN-affine updates) is a widely-adopted TTA approach that optimizes prediction entropy and BN statistics online [32]; related source-free and pseudo-labeling approaches (e.g., SHOT) adopt entropy / information-maximization objectives and feature clustering for target adaptation [18]. TTA methods can be very lightweight and effective across corrupted/shifted benchmarks, which is why they are attractive for per-client personalization in federated deployments. However, their online, label-free nature also changes the attack surface.

2.2. Adversarial attacks on test-time adaptation

TTA’s runtime updates open new avenues for adversaries. Classic data and model-poisoning literature (e.g., targeted clean-label poisoning, poisoning SVMs) established that training-time data manipulation can cause severe downstream errors [5, 26]. More recently, a host of works have analyzed and demonstrated test-time poisoning where an adversary injects malicious inputs during deployment to steer online adaptation. Notable attacks include the Distribution Invading Attack (DIA), which shows how inserting a small fraction of malicious examples into test batches can hijack BN-based and entropy-based TTA methods [34], and TePA, which performs surrogate-model-based test-time poisoning and demonstrates transferability across TTA variants [7]. Empirical studies show that only a few poisoned test samples can substantially degrade TTA performance on standard corrupted benchmarks [7, 34].

2.3. Federated Poisoning

Recent advancements in FL security indicate a transition from rudimentary data poisoning to persistent, multi-round model poisoning attacks that exploit temporal consistency to evade detection [36]. Advanced frameworks such as PoisonedFL [41] demonstrate that maintaining consistent malicious directions across training rounds can effectively neutralize many current Byzantine-robust aggregation methods. Zhou et al. [44] demonstrated that federated knowledge

graph embeddings are susceptible to poisoning through relation inference patterns that force the model to predict targeted false facts. Similarly, Yin et al. [38] introduced the PoisonFRS attack, which utilizes fake users to promote specific items in federated recommender systems without requiring prior knowledge of genuine user data. In the realm of mobile computing, Xiao et al. [35] explored perception poisoning attacks where fake nodes inject malicious updates to disrupt federated object detection learning. To mitigate these risks, Zheng et al. [43] proposed a defense using visual explanations and autoencoders to identify adversarial patterns via heat map reconstruction errors. Finally, Mai et al. [20] developed the Rflpa framework, which combines secure aggregation with cosine-similarity-based screening to robustly filter malicious gradients while maintaining user privacy.

3. Problem Setup and Threat Model

3.1. Federated Test-Time Adaptation

We consider a federated learning (FL) system with N clients coordinated by a central server. At round t , the server maintains a global model θ^t , which is broadcast to a subset of participating clients $S_t \subseteq \{1, \dots, N\}$. Each client $i \in S_t$ receives θ^t and adapts it to its local unlabeled stream $\mathcal{D}_{i,t}$ using a TTA algorithm T_i , such as entropy minimization [32], batch-normalization (BN) adaptation, or self-training methods [18]. After adaptation, the client returns either the updated model parameters θ_i^t or their difference $\Delta\theta_i^t$ to the server, which aggregates the received updates (e.g., via FedAvg) to produce the new global model:

$$\theta^{t+1} = \theta^t + \eta \sum_{i \in S_t} \frac{n_i}{\sum_{j \in S_t} n_j} \Delta\theta_i^t, \quad (1)$$

where n_i denotes the effective sample size for client i .

3.2. Adversarial Federated Client

Unlike prior poisoning works assuming white-box access to all updates or centralized validation sets, our attacker client operates poisoning attack during test time personalization in Federated Learning under a *realistic, constrained* setting:

- **Limited visibility:** the attacker observes only the broadcast global model θ^t at each round. It has no access to other clients' local data, their local updates, or the details of their adaptation trajectories.
- **Local control:** the attacker controls its own local data stream $\mathcal{D}_{a,t}$ with a fixed ratio and local adaptation process, but cannot access or interfere with other clients' data streams. It can inject poisoned examples $\mathcal{D}_{a,t}^p \subset \mathcal{D}_{a,t}$ or perturb existing inputs.
- **Stealth requirements:** the attacker's local updates must remain within norm bounds to avoid detection. Perturbations to inputs are constrained by ℓ_∞ or ℓ_2 budgets,

and label distributions must remain balanced to evade anomaly detection.

3.3. Attack Objective

The attacker aims to degrade the performance of benign clients after federated aggregation and subsequent test-time adaptation. Let $B_{i,t} = \{(x, y)\}$ denote a batch of samples from benign client i at round t , and let $T_i(\cdot)$ represent the TTA procedure used by client i . After the server aggregates client updates to obtain the global model θ^{t+1} , each benign client adapts it locally as $\theta_i^{t+1} = T_i(\theta^{t+1}, B_{i,t})$ and performs inference using the prediction function $h(x; \theta_i^{t+1})$.

The attacker controls a poisoned subset $\mathcal{D}_{a,t}^p$ within its local stream and seeks to maximize the expected loss of benign clients:

$$\max_{\mathcal{D}_{a,t}^p} \mathbb{E}_{i \neq a} [\mathcal{L}(h(x; \theta_i^{t+1}), y)], \quad (2)$$

where $\mathcal{L}(\cdot)$ denotes the classification loss. The objective is optimized under constraints on attacker participation, perturbation budgets, and update norms.

Because the attacker cannot observe benign batches $B_{i,t}$ or their adapted parameters θ_i^{t+1} , it approximates this objective using a local clean proxy set $\mathcal{B}_{ab}$ that mimics benign data distributions and a surrogate global model $\hat{\theta}^{t+1}$ estimated from historical global updates.

4. Methods

In this section we develop a practical attack framework for Realistic Federated Test-Time Poisoning (R-FTTP). Our approach extends single-node RTTDP ideas [27] to the federated setting by (a) building a lightweight surrogate aggregator that predicts the post-aggregation model available to honest clients, (b) crafting *in-distribution* poisoned examples whose intermediate-feature statistics are constrained to match a benign local pool, and (c) optimizing *TTA-aware* objectives that directly target common client adaptation mechanisms (e.g., BN adaptation, entropy minimization). We first give an overview, then present the surrogate aggregator, the feature-regularized poisoning formulation, the TTA-aware objectives, and finally an end-to-end optimization algorithm with practical constraints.

4.1. Overview

At each federated round t the server broadcasts the global model θ^t to a subset of clients S_t . Each client $i \in S_t$ performs local test-time adaptation T_i on its incoming unlabeled stream $\mathcal{D}_{i,t}$ and computes a local update $\Delta\theta_i^t$ that is returned to the server for aggregation. Our adversarial client a seeks to craft a poisoned local dataset $\mathcal{D}_{a,t}^p$ (a small fraction of its training stream) so that after aggregation and subsequent local TTA on other clients, honest clients suffer substantial degradations. Because the attacker cannot

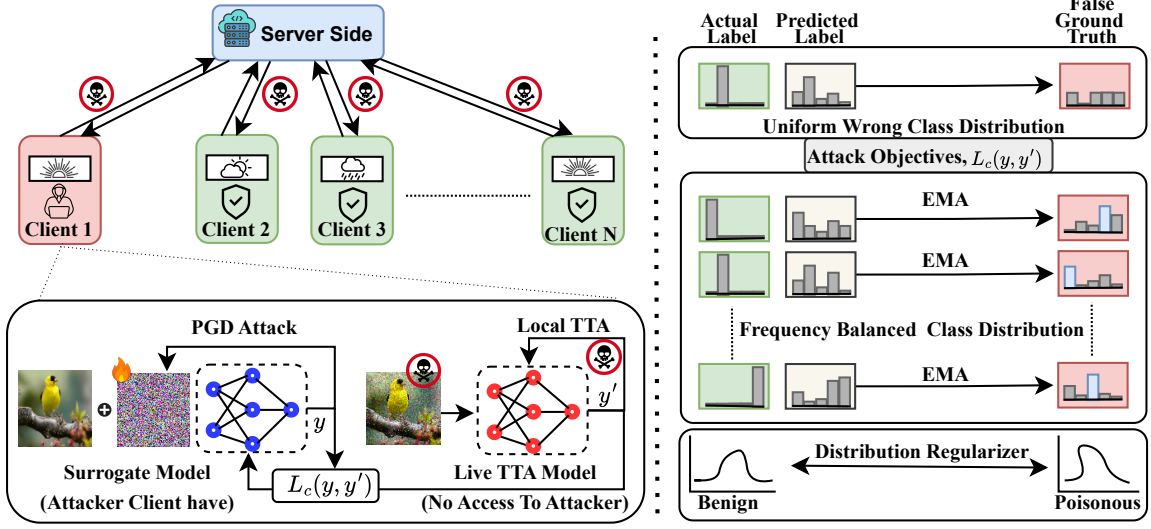


Figure 1. Overview of the FedPoisonTTP threat model. A subset of clients is compromised and injects poisoned inputs during local test-time adaptation. The attacker uses a surrogate model to craft in-distribution poisons via PGD-based optimization, which are then fed into the live TTA model, causing corrupted updates that propagate through the federated aggregation process and degrade the performance of honest clients.

access other clients' data or raw updates, it relies on (i) a *surrogate aggregator* $\hat{\theta}^{t+1}$ that approximates the true post-aggregation model, and (ii) a local benign pool $\mathcal{B}_{ab}$ to estimate cross-client transfer. The attack optimizes the poisoned inputs through projected gradient steps in order to maximize a surrogate outer loss evaluated on $\hat{\theta}^{t+1}$, while simultaneously enforcing a feature-distribution regularizer to preserve transferability and maintain stealth.

4.2. Surrogate Aggregator

The attacker cannot observe other clients' contributions to aggregation. We therefore define a surrogate aggregator that combines (A) a history-based estimate of other-clients' average update and (B) a posterior-distillation step that refines the surrogate using the broadcasted global model.

History-based aggregation estimate. The attacker stores the last k global models $\{\theta^{t-k+1}, \dots, \theta^t\}$ and forms a historical average update

$$\widehat{\Delta\theta}_{\text{hist}} = \frac{1}{k} \sum_{j=t-k+1}^t (\theta^j - \theta^{j-1}). \quad (3)$$

Assuming other clients' average update does not change rapidly, we estimate the other-clients' contribution at round t by

$$\widehat{\Delta\theta}_{-a} \approx \widehat{\Delta\theta}_{\text{hist}} - \widehat{\Delta\theta}_{a,\text{est}}, \quad (4)$$

where $\widehat{\Delta\theta}_{a,\text{est}}$ is the attacker's estimate of its own historical contribution (maintained by the attacker).

Candidate post-aggregation surrogate. Given a candidate local attacker update $\Delta\theta_a$ (obtained by training on a candidate poisoned set $\mathcal{D}_{a,t}^p$), the attacker constructs a surrogate post-aggregation model

$$\hat{\theta}^{t+1}(\Delta\theta_a) = \theta^t + \eta(\widehat{\Delta\theta}_{-a} + \Delta\theta_a), \quad (5)$$

where η is the aggregation scaling (server learning rate or weighting factor). Equation (5) is intentionally simple and inexpensive to compute; in practice the attacker refines $\widehat{\Delta\theta}_{-a}$ online by comparing its surrogate predictions to the actual broadcast θ^{t+1} once available (posterior distillation), i.e., minimizing a distillation loss on the attacker's benign pool $\mathcal{B}_{ab}$. Concretely, after receiving θ^{t+1} the attacker may update an internal state s by minimizing

$$\mathcal{L}_{\text{distill}}(s) = \frac{1}{|\mathcal{B}_{ab}|} \sum_{x \in \mathcal{B}_{ab}} \text{KL}(h(x; \theta^{t+1}) \| h(x; \hat{\theta}^{t+1}(s))), \quad (6)$$

where $h(\cdot; \theta)$ are model posteriors and s parameterizes the surrogate dynamics (e.g., corrections to $\widehat{\Delta\theta}_{-a}$). This posterior distillation improves future surrogate quality while respecting the attacker's limited visibility.

4.3. In-distribution Poisoning with Feature Regularizer

A key challenge is ensuring that poisoned examples generated on the attacker's local distribution transfer to unseen client domains. To address this, we constrain poisoning to remain *in-distribution* by matching the intermediate feature

statistics of the poisoned samples to those of a small local benign pool $\mathcal{B}_{ab}$, thereby maintaining alignment with the underlying data distribution.

Features and sample statistics. Let $f^l(x; \theta)$ denote the activation (flattened over spatial dimensions) at layer l for input x . For a sample x we compute its feature mean and (diagonal) variance at layer l :

$$\begin{aligned}\mu^l(x) &= \frac{1}{M_l} \sum_{m=1}^{M_l} f_m^l(x), \\ \sigma^l(x)^2 &= \frac{1}{M_l} \sum_{m=1}^{M_l} (f_m^l(x) - \mu^l(x))^2,\end{aligned}\quad (7)$$

where M_l is the number of flattened feature dimensions at layer l . For a set $\mathcal{S}$ we denote the empirical mean and variance by $\mu^l(\mathcal{S})$ and $\sigma^l(\mathcal{S})^2$.

Feature-distribution regularizer. For a candidate poisoned set $\mathcal{D}_{a,t}^p$ we define a layer-averaged moment-matching regularizer:

$$\begin{aligned}\mathcal{L}_{\text{reg}}(\mathcal{D}_{a,t}^p, \mathcal{B}_{ab}) &= \frac{1}{L} \sum_{l=1}^L \left(\|\mu^l(\mathcal{D}_{a,t}^p) - \mu^l(\mathcal{B}_{ab})\|_2^2 \right. \\ &\quad \left. + \beta \|\sigma^l(\mathcal{D}_{a,t}^p) - \sigma^l(\mathcal{B}_{ab})\|_2^2 \right),\end{aligned}\quad (8)$$

where L is the number of selected layers and $\beta > 0$ balances mean vs variance matching. Using diagonal moments keeps computation inexpensive and stable; in practice we select a small set of early- and mid-level layers (e.g., conv1, conv3, a penultimate block) that capture domain characteristics relevant to many clients.

4.4. TTA-aware Attack Objectives

We target two complementary TTA pathways commonly used by clients: entropy-minimization-style updates (e.g., TENT) and BN/statistics adaptation. Each objective is optimized while keeping $\mathcal{L}_{\text{reg}}$ small to ensure transferability.

BN-shift objective. BN-based TTA methods adapt to incoming batch statistics; by carefully shifting channel-wise means/variances the attacker can bias BN-affine parameters or running-statistics in the aggregated model. Let $g_{\text{BN}}(\theta, x)$ denote the vector of channel means/variances at selected early layers for input x . The BN-shift objective encourages a targeted shift s in channel statistics:

$$\mathcal{L}_{\text{BN}}(\mathcal{D}_{a,t}^p; \theta) = -\frac{1}{|\mathcal{D}_{a,t}^p|} \sum_{x \in \mathcal{D}_{a,t}^p} \langle g_{\text{BN}}(\theta, x), s \rangle, \quad (9)$$

where the sign is chosen so that local updates foster a persistent shift in the aggregated model’s BN-related parameters. The target shift s may be estimated from the attacker’s surrogate runs (for instance, directions known to degrade validation accuracy under simulated TTA).

Notch High Entropy (NHE) Attack Objective [27]: Self-training TTA methods such as TENT and RPL are highly vulnerable to high-entropy OOD inputs, motivating entropy-maximizing attacks such as TePA. However, maximizing entropy alone does not guarantee incorrect pseudo-labels, as the model may still place probability on the true class. To overcome this, we introduce the *Notch High Entropy (NHE) Attack*, which constructs a notched target distribution Q by setting $Q_y = 0$ and uniformly distributing mass across all incorrect classes. The corresponding loss is

$$\mathcal{L}_{\text{NHE}}(\mathcal{D}_{a,t}^p; \theta) = \frac{1}{|\mathcal{D}_{a,t}^p|} \sum_{x \in \mathcal{D}_{a,t}^p} \text{CE}(h(x; \theta), Q(x)), \quad (10)$$

which enforces high entropy while forcing the model toward systematically wrong predictions, thereby amplifying degradation during entropy-based TTA.

Balanced Low-Entropy (BLE) objective [27]: To attack entropy-minimization TTA, we construct class-balanced low-entropy targets that nudge the model toward confident but incorrect predictions across classes (avoiding class-collapse that is easily detected). Let $q(x)$ be a target pseudo-label distribution for x chosen such that the true class has near-zero mass and the remaining mass is distributed to non-true classes in a class-balanced fashion. The BLE loss on $\mathcal{B}$ under model θ is

$$\begin{aligned}\mathcal{L}_{\text{BLE}}(\mathcal{D}_{a,t}^p; \theta) &= \frac{1}{|\mathcal{D}_{a,t}^p|} \sum_{x \in \mathcal{D}_{a,t}^p} \text{CE}(h(x; \theta), q(x)) \\ &\quad + \gamma \mathcal{R}_{\text{conf}}(h(\cdot; \theta)),\end{aligned}\quad (11)$$

where CE is cross-entropy and $\mathcal{R}_{\text{conf}}$ is a regularizer that encourages global class balance of the induced confident labels (e.g., negative entropy of the empirical class-frequency vector) with weight γ . Minimizing (11) makes the model produce low-entropy but systematically wrong predictions after adaptation, which harms clients that use entropy minimization or pseudo-labeling.

5. Result & Discussion

5.1. Experimental Setup

Datasets: We conduct the experiments on two corruption benchmark datasets: CIFAR-10-C and CIFAR-100-C that have 15 different image corruptions (e.g., Gaussian noise,

Table 1. Comparison of federated learning (FL) methods with different test-time adaptation (TTA) techniques on CIFAR-10-C and CIFAR-100-C benchmark datasets with poison ratio, $\alpha = 0.5$. The total number of clients is considered to be $N = 10$, out of which 5 ($M = 5$) are considered to be malicious. The poison ratio indicates the ratio of samples that are corrupted in the batches of the malicious clients.

Fed Method	TTA Method	CIFAR-10-C			CIFAR-100-C		
		Clean	Attack Scenario		Clean	Attack Scenario	
			Benign Clients	Adversarial Clients		Benign Clients	Adversarial Clients
Any	Source	56.51	56.66	56.05	54.28	53.24	52.86
None	Tent	81.08	80.97	74.32	68.59	67.18	52.92
	CoTTA	81.78	78.45	69.21	66.24	61.73	48.15
FedAvg	Tent	81.19	77.95	75.43	68.13	55.15	56.09
	CoTTA	81.61	74.82	70.38	65.94	50.67	51.24
FedProx	Tent	81.20	78.17	75.62	68.13	55.22	56.11
	CoTTA	81.59	75.03	70.55	65.95	50.79	51.28
pFedGraph	Tent	81.19	77.85	75.49	68.11	55.16	55.98
	CoTTA	81.64	74.71	70.45	65.99	50.71	51.12
FedAmp	Tent	81.19	78.02	75.53	68.15	55.39	56.13
	CoTTA	81.61	74.95	70.51	65.95	50.94	51.31

blur, brightness) at five discrete severity levels from 1 to 5. In this work, we focus exclusively on the worst-case noise level: severity-5, thus simulating maximal domain shift to stress-test the model’s robustness.

Implementation details: We utilize the WideResNet-28 and ResNeXt-29 models for CIFAR-10-C and CIFAR-100-C datasets, respectively. We test the robustness of two popular Test-Time Adaptation (TTA) methods: TENT [31] and COTTA [33] under four different federated aggregation strategies: FedAvg [21], Fedprox [15], pFedGraph [37], and FedAMP [11]. The batch size for each client is chosen to be 100. We perform the experiments for a total of $N = 10$ clients, of which $M = 5$ are adversarial, while the remaining 5 are benign. We introduce the term, poison ratio, α , to indicate the ratio of samples within the adversarial clients’ batch which are perturbed. For our experiments poison ratio (α) is considered to be 0.5 for the adversarial clients.

5.2. Results

We evaluate our framework, FedPoisonTTP, across CIFAR-10-C and CIFAR-100-C benchmarks under diverse configurations of federated and test-time adaptation (TTA) setups to analyze the impact of adversarial clients on collaborative personalization and robustness. We report the results of our experiments in Table 1. Here, ‘Source’ indicates no Test-Time adaptation, where the evaluation scores of the frozen models on the test-sets are reported. While ‘None’ indicates no federated aggregation, where the average accuracy values for each participating client (benign samples) are reported for various TTA techniques.

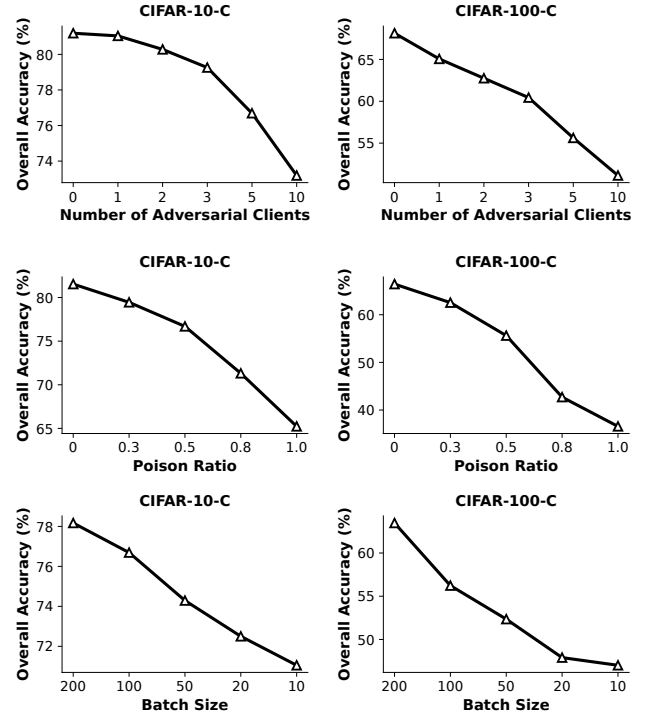


Figure 2. Impact on overall accuracy for FedAvg-TENT setup due to varying the number of adversarial clients, ratio of malicious samples (α), and the batch size for the participating clients. Overall accuracy reflects the average performance across the test sets of benign clients as well as the benign portion of the adversarial clients’ test data.

Degradation of Federated TTA under Poisoning Attacks. Table 1 reports the performance of four federated learning methods: FedAvg, FedProx, pFedGraph,

Table 2. Performance comparison under different poisoning objectives on CIFAR-10-C and CIFAR-100-C corruption benchmarks for FedAvg-TENT setup. Under ‘Benign’ and ‘Adversarial’, we report the accuracies of the benign clients and the benign data partition of the adversarial clients. ‘Overall’ is the average accuracy of the benign clients and adversarial client’s benign partition.

Poisoning Objective	CIFAR-10-C						CIFAR-100-C					
	White box Attack			Grey box Attack			White box Attack			Grey box Attack		
	Overall	Benign	Adversarial	Overall	Benign	Adversarial	Overall	Benign	Adversarial	Overall	Benign	Adversarial
NHE	73.47	77.44	69.50	74.09	77.47	70.71	55.57	58.60	52.54	55.24	58.30	52.18
NHE + Distribution Regularization	76.25	77.41	75.10	76.69	77.95	75.43	55.42	54.58	56.26	55.62	55.15	56.09
BLE	74.20	80.81	67.58	74.54	80.84	68.23	62.48	66.63	58.33	63.04	66.80	59.29
BLE + Distribution Regularization	78.14	80.19	68.23	78.55	80.40	76.70	65.68	66.92	64.30	66.10	67.29	64.92
MaxCE	77.62	80.97	74.26	77.54	81.02	74.06	63.36	66.58	60.15	63.49	66.64	60.35
Tepa	77.68	80.85	74.51	75.24	78.99	71.49	54.07	56.35	51.78	54.32	56.86	51.77
DIA	78.56	81.06	76.07	78.28	81.03	75.52	62.75	65.91	59.60	62.59	65.91	59.28
Distribution Regularization	79.62	81.31	77.94	79.66	81.38	77.95	64.21	65.05	63.37	64.25	65.08	63.43

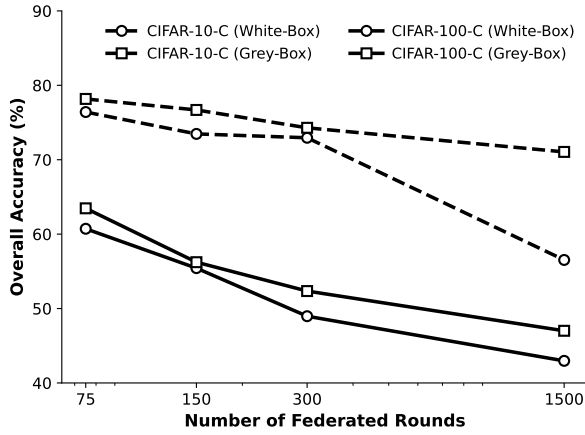


Figure 3. Overall Accuracy (%) vs. Number of Federated Rounds. Performance on the benign clients and the benign partition of the adversarial clients (overall accuracy) on CIFAR-10-C (dashed lines) and CIFAR-100-C (solid lines) under White-Box (black) and Grey-Box (orange) attacks.

and FedAMP, combined with the aforementioned test-time adaptation schemes: TENT and COTTA. Here, ‘clean’ indicates the case of no adversarial clients, i.e., all clients are benign. For the attack scenario, we show the accuracies for the benign clients and the benign partition (50% data in the batch) in the test set for the adversarial clients. As expected, in clean conditions, both TENT and COTTA yield strong accuracies (around 81% on the CIFAR-10-C and around 68% on CIFAR-100-C) for all the Fed-TTA setups. When five out of ten clients become adversarial and poison half (poison ratio, $\alpha = 0.5$) of their local batches, the test-time accuracy across all settings is significantly reduced. For instance, adaptation via TENT under FedAvg aggregation has a performance drop from 81.19% to 77.95% on CIFAR-10-C and 68.13% to 55.15% on CIFAR-100-C, respectively, for the benign clients. The federated adaptations under COTTA show significant degradation in performance due to the participation of adversarial clients, as this strategy up-

dates a higher number of parameters compared to TENT. For FedAvg-COTTA, we see a drop of 6.79% accuracy. While in the same configuration, for CIFAR-100-C we observe that the presence of adversarial clients is even more detrimental, with performance dropping by 14.47% for the benign clients.

Impact Across Datasets. FedPoisonTTP demonstrates performance degradation for both CIFAR-10-C & CIFAR-100-C datasets. However, the performance drop is more prominent on CIFAR-100-C due to its higher class granularity and sensitivity to domain shifts. On average, FedPoisonTTP decreases benign clients’ accuracy by 3.03–3.34% on CIFAR-10-C, and by 12.76–12.98% for CIFAR-100-C under TENT adaptation. The pronounced degradation on CIFAR-100-C underscores the practical impact of FedPoisonTTP, as the dataset’s larger number of classes and finer-grained semantic structure better reflect real-world scenarios where models must generalize across diverse domains.

5.3. Ablation Studies

We further analyze the effect of varying the number of adversarial clients, poison ratio (α), and batch size on the overall performance of the Test-time setup. The overall accuracy, in this case, is the average accuracy of the test set of the benign clients and the benign partition of the adversarial clients. Figure 2 shows the impact of varying the number of adversarial clients, poison ratio, and batch size on the overall accuracy for the FedAvg-TENT setup.

Effect of Adversarial Participation. Figure 2 (top) shows how the overall accuracy varies with the increasing number of adversarial clients for a fixed poison ratio of $\alpha=0.5$. The performance on CIFAR-10-C falls from 81.19% to 73.19% as the number of adversarial clients increases. As discussed earlier, due to having a higher number of classes, this impact is more noticeable on the CIFAR-100-C dataset where the accuracy falls from 68.13% to 51.13%, yielding a performance decline of 17%. This demonstrates that even

a single compromised client can cause a noticeable drop in performance, underscoring the susceptibility of Fed-TTA to localized poisoning attacks.

Effect of Poison Ratio. Figure 2 (middle) depicts the implications of increasing the poison ratio (α), keeping the number of malicious clients (M) fixed to 5. We observe a drop of around 17.34% and 29.86% accuracy values, compared to the clean scenario, on the CIFAR-10-C and CIFAR-100-C benchmarks, respectively. This implies that increasing the amount of perturbed samples across the malicious clients is significantly detrimental to the collaborative adaptation scheme. Therefore, compared to increasing the number of adversarial clients, increasing the number of poisoned samples can significantly deteriorate the performance of the benign clients.

Effect of batch size Figure 2 (bottom) reports the impact of varying the batch size for all clients while keeping the number of clients fixed (10 in total; 5 of them are adversarial) and using a poison ratio of 0.5. We vary the per-client batch size from 200 down to 10 and evaluate the federated adaptation performance on the CIFAR-10-C and CIFAR-100-C datasets. As the batch size decreases, the accuracy shows a declining trend for both datasets. This trend arises because smaller batches render each client’s update more sensitive to the poisoned samples injected by the adversarial participants. When the batch size is large, the poisoned portion (50% of the batch for malicious clients) is diluted by a greater number of overall samples, causing the adversarial influence to be significantly reduced. For instance, on both CIFAR-10-C & CIFAR-100-C datasets we see high accuracy values of 78.17% and 63.47%, respectively, for a batch size of 200. Conversely, for a batch size of 10, the performances on both CIFAR-10-C & CIFAR-100-C drop to 71.05% and 47.01% values, respectively, with the adversarial effect being more prominent on CIFAR-100-C. Therefore, with small batches, each poisoned example contributes a larger proportion of the adaptation signal, amplifying the disruption caused during test-time updates.

Analysing impact of attacks over various Federated Rounds In Figure 3, we report the overall classification accuracy with respect to the varying number of federated rounds. The batch size per client is varied over the values: 10, 50, 100, and 200, which correspond to the number of federated rounds: 1500, 300, 150, and 75. A reduction in the local batch size increases the total number of federated rounds, thereby providing additional opportunities for the adversarial updates to propagate through repeated model aggregation. This effect appears consistently across both datasets. Due to having a higher number of classes, this effect is more prominent for the CIFAR-100-C dataset.

Analyzing the effects of various poisoning objectives In Table 2, we analyze the impact of several poisoning objectives under both white-box and grey-box (surrogate model) settings on CIFAR-10-C and CIFAR-100-C. We adopt the FedAvg-TENT setup with 5 malicious clients, a poison ratio of 0.5, and a batch size of 100 per client. A consistent trend is observed where white-box attacks lead to stronger degradation than grey-box attacks, as direct access to model parameters enables more precise gradient-based manipulation. Across both datasets, objectives such as DIA, MaxCE, and Tapa significantly reduce adversarial-partition accuracy while maintaining relatively higher benign accuracy, indicating limited propagation of poisoned effects to benign clients. In contrast, the NHE objective results in a substantial performance drop for benign clients. On the more challenging CIFAR-100-C benchmark, all attacks are more pronounced, with lower overall accuracy across all objectives and settings, reflecting the increased sensitivity of this higher-class dataset to perturbations during adaptation. Furthermore, combining NHE with distribution regularization enhances the attack impact on CIFAR-100-C due to its more fine-grained class distinctions.

6. Conclusion

This work introduces FedPoisonTTP, a realistic grey-box poisoning framework designed to examine the vulnerability of federated test-time personalization. By modeling an adversarial client with limited visibility and constrained poisoning capabilities, we highlight how test-time adaptation, particularly entropy-based methods, can be exploited to propagate harmful updates across collaborative learning rounds. Our empirical benchmark, spanning multiple federated aggregation methods combined with online test-time adaptation, demonstrates that even constrained adversarial participants can induce consistent and significant reductions in post-adaptation accuracy. Through extensive ablations, we further isolate the key factors that influence the severity of the attack, including the number of adversarial clients, poisoning ratio, and batch size. Overall, our findings underscore a fundamental and concerning security gap in federated test-time adaptation, while also providing a solid foundation and valuable insights for future research aimed at developing principled defenses, enhanced robustness, and more resilient training procedures.

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